A Conceptual Introduction to SCTP

Let's see what is SCTP, and its advantages over TCP.

The Stream Control Transmission Protocol (SCTP) is an IP transport protocol. It resides at an equivalent level with TCP, and provides a reliable transport service, ensuring that data is transported across the network without error, and in sequence. It is designed to transport Public Switched Telephone Network (PSTN) signalling messages over IP networks, but is capable of broader applications.

SCTP employs a session-oriented mechanism (similar to TCP) for transmission of data. A relationship is established between the endpoints of SCTP associations prior to data transfer, and this relationship is maintained until all data transmission has been successfully completed.

There are many good features in SCTP, not all of which, obviously, can be covered in one article. The most important ones are:

1. Multi-stream capability
2. A broader concept for connecting endpoints (similar to, and more useful than, TCP connection), known as 'association'
3. Multi-homing support
4. Path and peer failure detection

Let's look at these in detail.

Why the name 'Stream'?

What's in a name? A lot, in this case. While TCP assumes a single-stream data transfer with byte sequence preservation, SCTP offers the multi-streaming facility. The term 'stream' in SCTP means "...a sequence of user messages that are to be delivered to the upper-layer protocol in order, with respect to other messages within the same stream." This is in contrast to its use in TCP, where it refers to a sequence of bytes.

Streams allow user data to be partitioned into multiple 'streams', each with independently sequenced delivery. Therefore, message loss at one stream will not affect (or hold up) the delivery of other